

Overlap Integral Analysis for Second-Harmonic Generation Within Inverted Waveguide Using Mode Dispersion Phase Match

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Abstract—We have discussed the design of a multilayer waveguide toward an efficient nonlinear interaction. We have investigated the five- and seven-layer inverted waveguide. The effect of the interlayer thickness and refractive index on the waveguide overlap integral and phase match thickness has been described. A simple approach for improving the nonlinear interaction within inverted waveguide through maximization of the overlap integral is presented.

Index Terms—Inverted waveguide, mode dispersion, nonlinear optical devices, optical waveguide, phase matching.

I. INTRODUCTION

AMONG THE nonlinear optical effects the widely used and investigated is the second-harmonic generation (SHG). The SHG process is used not only to generate new frequency but via nonlinear cascading to induce an intensity-dependent refractive index.

Molecular nonlinear optics has matured over the last two decades into an active domain of applicative research with a lot of promising implication into optical communication. Breakthroughs have been reported for electrooptical polymers [1], which show electrooptical coefficient higher than that of inorganic crystals. These new developments have gradually cleared misconceptions about polymers being fragile and unstable materials.

In this letter, we address the design issues of an organic inverted waveguide for efficient SHG using mode dispersion phase match (PM).

While the energy is converted back and forth between the fundamental and second harmonic, the fundamental wave undergoes a nonlinear phase shift. If the interaction is taking place within nonlinear waveguide, then the efficiency is high enough to be used for application. A recently reported value [2] for intensity-dependent refractive index within organic waveguide due to second-order cascading effect is $n_2 = 2.68 \cdot 10^{-13} \text{ cm}^2/\text{W}$, which is three orders of magnitude higher than the corresponding parameter due to third-order nonlinear effect ($n_2 = 7.9 \cdot 10^{-16} \text{ cm}^2/\text{W}$) on the same organic material [3].

For an efficient nonlinear interaction within waveguide, the structure should fulfill two requirements.

First, the waveguide should be designed to support the fundamental and second-harmonic waves as well as both should be phase matched. In a normal dispersion waveguide, the $n_{\text{eff}}(2\omega) > n_{\text{eff}}(\omega)$, so the fundamental and SH wave travel at different velocities. The PM is achieved by selecting a proper waveguide mode dispersion. By the thickness and the refractive index of waveguide material, the effective refractive index of different modes are tuned to reach PM. In this process, the waveguide thickness is critical, which makes it difficult from a technological point of view. This is called the mode dispersion PM method. Modal dispersion PM is attractive for materials with large nondependent diagonal nonlinear susceptibilities as polymers. The drawback is the optimization of overlap integral that should be addressed with extensive technological difficulty.

The second requirement is maximization of overlap integral. This is defined as

$$S = \frac{\left| \iint E_{\omega}^2 \cdot E_{2\omega} \cdot dx \cdot dy \right|}{\left(\iint E_{\omega}^2 \cdot dx \cdot dy \right) \cdot \left(\iint E_{2\omega}^2 \cdot dx \cdot dy \right)^{1/2}}$$

where E_{ω} and $E_{2\omega}$ is the electric field distribution of the fundamental and the second-harmonic wave. This formula is used for channel waveguide and has $(\text{length})^{-1}$ as the unit of measure. For the geometry considered here, the propagation direction is the z axis and the refractive index profile changes along the x axis. For the case of slab waveguide, the result of overlap integral along the y axis is trivial. Several years ago, the concept of reversing the sign of $\chi^{(2)}$ at the thickness where the electric field distribution of the first-order mode undergoes a sign reversal was proposed. If the product of the electric field distribution changes across the waveguide, then the value of the overlap is strongly reduced. By reversing the negative part of the field distribution for the SH wave, the maximum overlap integral between the fundamental and SH wave is present. This structure is called the inverted waveguide. Recently, a very simple technological approach was proved for realizing of an inverted waveguide with a power-corrected efficiency of 8.9% [2], [4], the highest reported within organic waveguide. Having prepared two identical well-organized organic films on quartz glass substrates, both films are pressed toward each other, forming a waveguide of twice the thickness of the initial organic film. The

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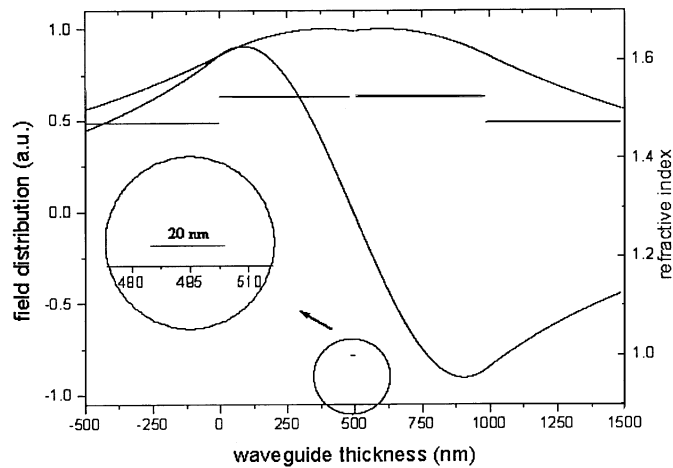


Fig. 1. Refractive index profile of a five-layer waveguide. In the present case, the air interlayer has 20-nm thickness. The electric field distribution for the TE mode of the first order for fundamental and second order for second harmonic is plotted. The core region has 2×490 nm.

net dipole of both organic films should be parallel but it should point opposite sense. The technological difficulty is ensuring an optical contact between the two films that make up the waveguide.

Advanced poling techniques allow for the design of optical devices having net dipole either vertical or horizontal. The vertical dipole orientation is achieved by corona or electrode poling technique, and horizontal orientation of organic film net dipole can be achieved by horizontal electrode poling. Each of these geometries imposed restriction on the waveguide mode used for nonlinear interaction. So if the net dipole of the polymeric film is vertical, then only the transverse magnetic (TM) mode interaction can be optimized with overlap integral. If the net dipole is in-plane oriented, then the overlap integral can be optimized for the transverse electric (TE) mode.

In Fig. 1, the refractive index profile of the waveguide is presented as well as the electric field distribution for the TE mode of the fundamental and second harmonic for having an air interlayer of 20-nm thickness. The refractive index profile is complex and is similar to that of a total internal reflection waveguide or an M-waveguide [5]–[8].

The question is how PM thickness and overlap integral change because of the interlayer thickness and refractive index variation. Three cases are going to be discussed that correspond to different materials for the interlayer: a) having an air interlayer; b) having as interlayer a material (liquid) with lower refractive index than the core region of the waveguide; and c) a material with higher refractive index than the core region of the waveguide.

In the waveguide design, we have used for the refractive indices values presented in Table I. The refractive index for cladding and substrate correspond to SiO_2 and for the core region corresponds to guest-host system of Polymethyl methacrylate & Disperse Red (PMMA&DR).

First, the waveguide thickness necessary to fulfill the PM condition is determined. Second, for the previous condition, the electric field distribution, and overlap integral is calculated. For easy comparison, we normalized the overlap integral and the

TABLE I
THE REFRACTIVE INDICES OF SUBSTRATE, CLADDING, CORE REGION OF WAVEGUIDE, AND INTERLAYER USED IN WAVEGUIDE DESIGN

Wavelength (nm)	Substrate & cladding	Core region	Interlayer		
			Air (case a)	Low (case b)	High (case c)
1550	1.4715	1.526	1.0	1.5	1.54
775	1.4775	1.56	1.0	1.53	1.57

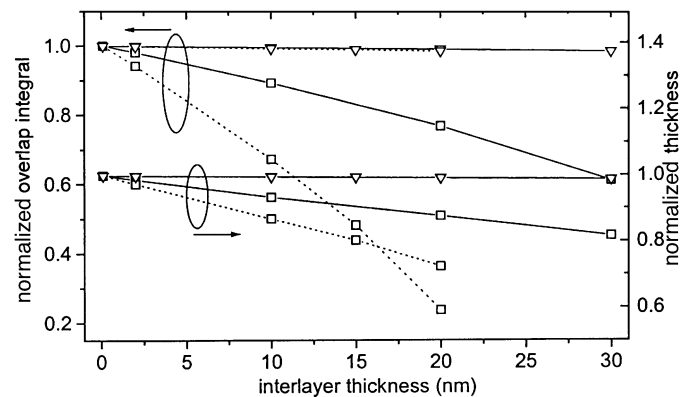


Fig. 2. Overlap integral and the PM thickness versus interlayer thickness for the case of having interlayer of air (square) and with the following refractive index 1.5 at 1550 nm (1.53 at 775 nm) (down triangle). The TE mode (full line) yields to a higher overlap integral than TM mode (dotted line).

waveguide thickness to the value of having no interlayer in between core layer of the waveguide. The PM is taking place between the zeroth-order mode of the fundamental and first-order mode of the second-harmonic wave.

The air interlayer in between the two core organic films yields a pronounced decrease of the overlap integral (Fig. 2). The PM thickness is reduced with increasing thickness of air interlayer. The decrease of overlap integral is more pronounced for the TM modes. For an interlayer thickness of 20 nm, the use of case (b) and an air-made interlayer yields a shift of the PM thickness toward a lower value with 1% and 13%, respectively, and a decrease of the overlap integral with 2% and 24%, respectively.

If the refractive index of interlayer film is higher than the refractive index of core films, then with the increasing of the interlayer thickness, the overlap integral is increased as well as the PM thickness. This approach can be used for improving the efficiency of SHG. As is presented in Fig. 3, the overlap integral corresponding to the TM mode is higher than for the TE mode. The maximum of the overlap integral is increased with 2% for 130-nm interlayer thickness for the case (c). If the refractive index of interlayer is increased to 1.6 at 1550 nm (1.63 at 775 nm), the maximum overlap integral is shifted toward thicker interlayer thickness, increasing with 28% for 280 nm. After this value of interlayer thickness, we notice a sharp decrease of the overlap integral.

It is known that a thin (satellite) layer with high refractive index near to the waveguide will lift up the electric field distribution and in some appropriate design can yield an increase of the overlap integral. The next structure investigated is a symmetrical seven-layer waveguide presented in Fig. 4. As can be seen

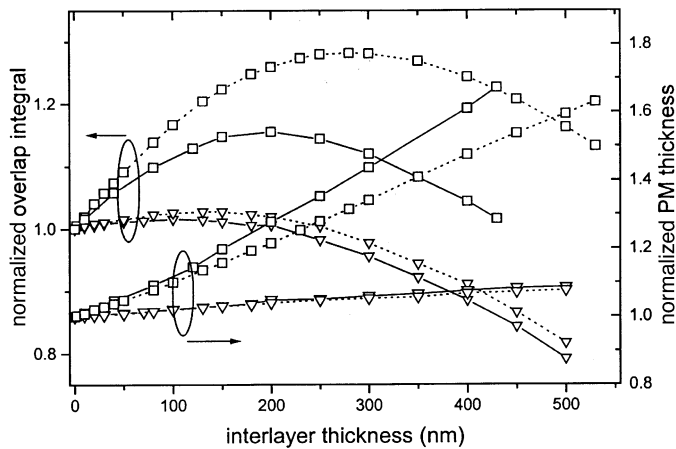


Fig. 3. The overlap integral and the PM thickness versus interlayer thickness. The down triangle corresponds to an interlayer having 1.54 at 1550 nm (1.57 at 775 nm), and square to the case of 1.6 at 1550 nm (1.63 at 775 nm). The TM mode (dotted line) yields to a higher overlap integral than TE mode (full line).

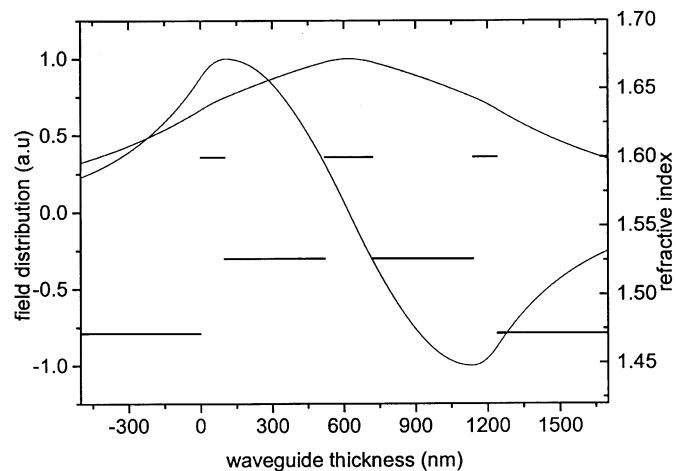


Fig. 4. Refractive index profile of the seven-layer waveguide. The electric field distribution of the zeroth-order mode of the fundamental and the first-order mode of the second-harmonic wave. The thickness of the satellite layer is 100 nm and the thickness of the interlayer is 200 nm. The core region has 2×420 nm.

in Fig. 4, the electric field distribution is lifted up by the presence of the satellite layer with higher refractive index than core region. The case of the satellite layer having lower refractive index than the refractive index of core region leads to lower value for the overlap integral than for case of five-layer waveguide. We have used for the interlayer and the satellite layer the same refractive index of 1.6 at 1550 nm (1.63 at 775 nm). The increase of the satellite-layer thickness yields a decrease of overlap integral. The results are presented in Fig. 5. In this case, we used as normalization value for overlap integral and PM thickness the maximum value of five-layer waveguide reached for 280-nm interlayer thickness with 1.6 at 1550 nm (1.63 at 775 nm).

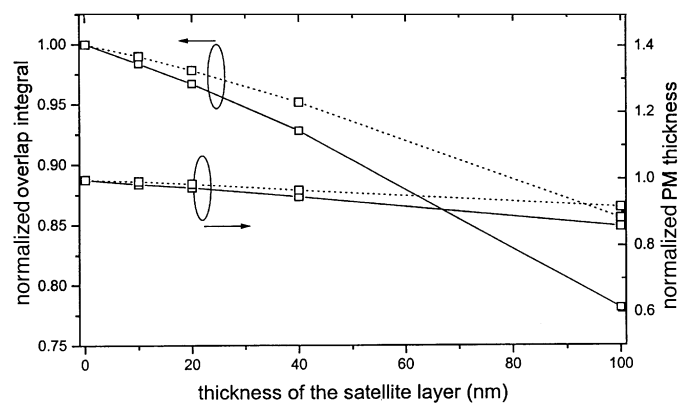


Fig. 5. Normalized overlap integral and PM thickness versus thickness of the satellite layer. The TM mode (dotted line) yields to a higher overlap integral than TE mode (full line).

The waveguide with maximum overlap integral is the one with five-layer structure. The effect of different refractive index and interlayer thickness on PM thickness and on overlap integral has been discussed. The use of an interlayer with different refractive index yields an increased flexibility of a multilayer waveguide structure, which allows to tune the PM thickness and optimize overlap integral.

We have described a simple algorithm for designing a waveguide with improved SHG efficiency interaction via increasing of the overlap integral.

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